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A LLAMA3-8B CIRCUIT EVALUATION RESULTS

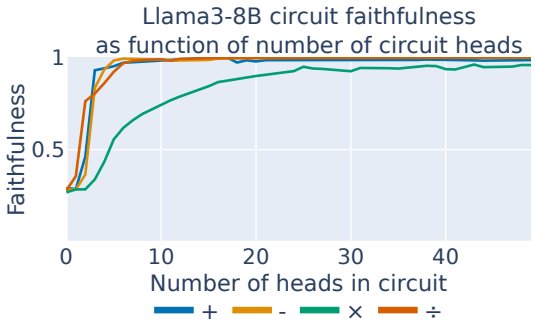


Figure 11: Llama3-8B arithmetic circuit faithfulness as function of number of circuit heads.

In Figure 11, we present an analysis of the faithfulness of the identified circuit in Llama3-8B (Section 2.1) as a function of the number of attention heads within the circuit. The circuit includes all MLPs (the neuron-level analysis in Section 3.1 does not apply in this context). Our goal is to find a minimal subset of the model with the fewest attention heads possible while maintaining high faithfulness. We assess faithfulness independently for each operator to obtain a more nuanced understanding of the necessary attention heads. For addition, subtraction, and division operations, we observe that 6 heads suffice to attain high faithfulness (97% on average). In contrast, in multiplication, 20 attention heads are required to achieve a faithfulness score exceeding 90%. The faithfulness of the full arithmetic circuit for each operator, corresponding to the Pareto-optimal number of heads (in terms of achieving high faithfulness with as few heads as possible), is documented in Table 1. We explore the attention patterns of these attention heads in the next section.

Table 1: Llama3-8B arithmetic circuit faithfulness, per operator.

Operator	+	-	×	÷
Faithfulness	0.97	0.98	0.90	0.96
# Attn Heads	6	6	20	6

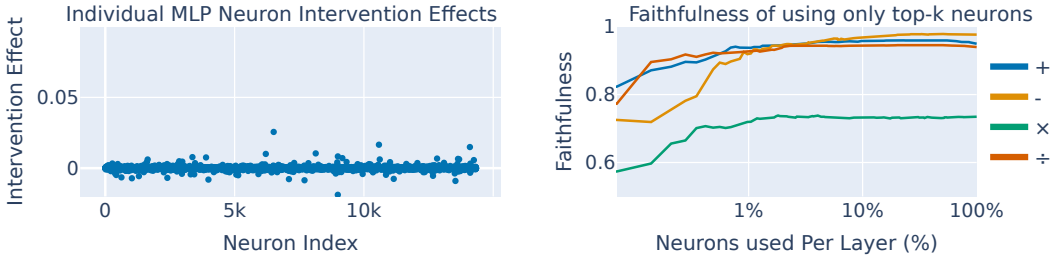
B LLAMA3-8B CIRCUIT ADDITIONAL COMPONENTS

To provide a more comprehensive understanding of the arithmetic circuit in Llama3-8B, we analyze the additional components that compose it, namely the first MLP layer (MLP0) and the high-effect attention heads.

B.1 MLP0

To analyze the role of MLP0 in the arithmetic circuit, we first test if—similarly to the middle- and late-layer MLPs—only a sparse set of neurons within the MLP is required. We measure the intervention effect of each neuron, averaged across the operands and operator positions ($p \in [1, 2, 3]$). The results, shown in Figure 12a, reveal that few MLP0 neurons have a high effect, similar to the middle- and late-layer MLPs. To verify these neurons are sufficient for arithmetic calculations, we repeat the experiment from Section 3.1. Specifically, we measure the faithfulness of the circuit—consisting of the top 1% of neurons in each of the middle and late layers ($l \in [16, 32]$) as well as a varying number of neurons in MLP0. The results, shown in Figure 12b, reveal that also in the first MLP layer, as little as 1% of neurons is sufficient for the circuit to achieve high faithfulness, similarly to the middle- and late-layer MLPs.

Because Llama3-8B applies a positional embedding only at attention layers, the activation of MLP0 is not affected by the position of any token. Additionally, due to a lack of attention heads that move information between the operand and operator positions before MLP0, we can analyze its effect directly on single tokens. Thus, we view MLP0 as an "effective embedding" (McDougall et al.,

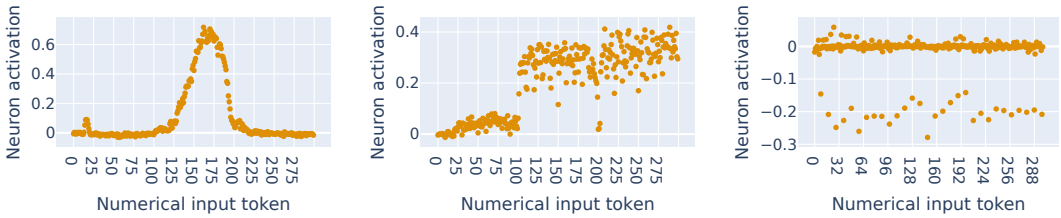


(a) Few neurons in the first MLP have a high effect on arithmetic prompts.

(b) The circuit faithfulness when including a percentage of high-effect MLP0 neurons in the circuit.

Figure 12: Analyzing effect of individual MLP0 neurons. A small number of neurons is sufficient for circuit accuracy.

2023), and hypothesize that the role of each MLP0 neuron is to incorporate additional numerical information into each token embedding. To verify this, we pass a list of numerical tokens, each representing an operand in our analyzed operand range ($t \in [0, 300]$), through the model. We measure the activation of each high-effect neuron for each token. As exemplified in Figure 13, the activations of these high-effect neurons correspond to varied numerical features. For example, Neuron 6206 (Figure 13a activates for numbers near 170 or 17; Neuron 7101 activates for numbers greater than 100; Neuron 8969 activates for numbers that are congruent to $8 \pmod{10}$. Overall, the set of patterns identified by these neurons can be used by the middle- and late-layer heuristic neurons to perform their more complex functionalities.



(a) Neuron 6206

(b) Neuron 7101

(c) Neuron 8969

Figure 13: Individual MLP0 neurons have identifiable activation patterns for numerical tokens.

B.2 Attention Head Patterns

We present the attention patterns of the attention heads that are contained in the arithmetic circuit. In Section 2.1, for simplicity, we consider the arithmetic circuit as a single circuit for all arithmetic operators, containing all attention heads used across the four operators. We observe that to achieve high faithfulness of the circuit with a minimal number of heads (Appendix A), some general arithmetic heads, that significantly improve faithfulness across all operators, are required. Other heads, while contributing to increased faithfulness, are operator-specific. They exhibit varying levels of importance for different operators. A full description of the attention heads used in the arithmetic circuit for each operator is provided in Table 2.

To better understand the role of the general arithmetic heads (L2H2, L15H13, L16H21), we compute their attention patterns, averaging them across our prompt dataset (Section 2.1). These patterns (Figure 14) reveal a clear signal: each of the three heads attends to a single input token, copying the representation from that position and projecting it to the last position. Specifically, L16H21 attends to the first operand, L2H2 attends to the operator and L15H13 attends to the second operand. This implies the role of each such head is to move the representation from that position, which includes to the last position, where it is further processed by the bag of heuristics implemented in the middle- and late-layer MLPs. To confirm that the information copied from each position to the last position consists solely of data from the token at that position, we conduct an ablation study. For each general arithmetic head, we zero out all preceding attention patterns that move information